Phase 1: Capitalist Baseline Simulation State Report

1. Overview

This simulation represents Phase 1 of the "ML Socialism" project. It is an Agent-Based Model (ABM) designed to simulate a decentralized, natural-selection-driven capitalist economy. The purpose of this phase is to establish a control group where populations interact, compete, and evolve based strictly on local information and localized "survival of the fittest" pressures, without any central planning or global optimization.

2. Environmental Architecture

The environment is a grid-based procedural map containing static entities called `ResourceCell` agents.

- **Map Generation:** The terrain is generated using randomized noise arrays subjected to multiple smoothing passes (default 5 passes for terrain, 7 for resources) to create contiguous continents and oceans.
- **Grid Structure:** The world distinguishes between "Land" and "Water" tiles. Land tiles are assigned a floating-point `resource_value` between 0.0 and 1.0.
- **Carrying Capacity (K):** Each land tile has a strict carrying capacity derived directly from its `resource_value`. By default, this capacity scales between a minimum of 90.0 and a maximum of 520.0 individuals per tile.

3. The Population Agent & Investment Tree

The primary actors in the simulation are `Population` agents. Each agent represents the entirety of the inhabitants on a single land tile, possessing a unified set of beliefs and a unique `lineage_color` to track ancestral nations.

A population's behavior is dictated by a strict mathematical investment tree that always normalizes to 1.0 (100%):

1. **Top-Level Abstract Split:**
   - `x_tech`: Allocation for technological advancement (max 30%).
   - `y_dip`: Allocation for diplomacy and trade leverage (max 30%).
   - `z_physical`: The remaining physical budget, calculated strictly as `1.0 - (x_tech + y_dip)`.
2. **The Guns vs. Butter Sub-Split:**

- `e_econ_ratio` : A slider (0.0 to 1.0) determining how the `z_physical` pool is spent.
- **Economic Proportion:** `z_physical * e_econ_ratio`
- **Military Proportion:** `z_physical * (1.0 - e_econ_ratio)`

The Technology Multiplier: Accumulated technology yields a `tech_level` (an integer). The simulation calculates a `tech_multiplier` using the formula `1.0 + (tech_level * 0.05)` . This multiplier is applied to the raw proportional outputs to calculate the final effective `military_output()` , `economic_output()` , and `diplomatic_output()` .

4. Internal Dynamics

Populations execute internal domestic actions on every tick before interacting with neighbors.

- **Dynamic Population Growth:** Population scales logistically based on the Euler integration formula: `dP/dt = r * P * (1 - P/K)` .
  - The base growth rate `r` (default 0.085) is positively modified by the economy: `economic_bonus = 1.0 + (economic_output * 0.05)` .
- **Generational Trait Drift:** On every tick, the core beliefs (`x_tech` , `y_dip` , `e_econ_ratio`) undergo a random normal mutation (`mean=0.0, std=0.01`). This simulates generations shifting their cultural/economic values over time.
- **Tech Diffusion:** Populations bordering richer, higher-tech neighbors passively absorb technological progress based on the gap between their levels and a `tech_diffusion_rate` (default 0.35).

5. Competition & Conflict Mechanics

The simulation features a robust, multi-step combat system designed to heavily penalize over-extension and reward diplomatic assimilation.

Step A: The Attack Trigger A population next to a foreign lineage evaluates whether to attack based on its internal population pressure.

- *Win Probability Assessment:* The attacker calculates an `actual_win_prob` by assessing the defender's total power (`defender_pop * defender_military_output`).
- *Deterrence:* A `dip_factor` is calculated as `max(0.1, defender_dip - attacker_dip + 1.0)` .
- *Probability Formula:* The chance to trigger an attack is `(actual_win_prob / dip_factor) * (population / K) * 0.05` .

Step B: The Smart Expeditionary Force Populations do not risk their entire demographic base. They calculate the minimum force required to achieve an 80% combat advantage:

- *Optimal Force:* `4.0 * (defender_power / attacker_military_output)` .

- *Capped Force:* This expedition is strictly capped at a maximum of 80% of the attacker's current total population (`min(optimal_force, pop * 0.8)`).

Step C: Combat Resolution & Assimilation

- A random roll against the `actual_win_prob` determines victory.
- *Attacker Loses:* The entire expeditionary force is permanently removed from the simulation.
- *Attacker Wins:* The target tile's lineage and traits are overwritten. The expeditionary force moves into the tile.
- *Assimilation Calculation:* The attacker absorbs a percentage of the surviving defenders based on their diplomatic investment: `min(1.25 * attacker_diplomatic_output, 1.0)` . Un-assimilated defenders are removed.

6. Expansion & Trade

- **Trade Arbitrage:** Adjacent populations with stockpiles can trade. The `diplomatic_delta` acts as a perceived value modifier up to a maximum 20% "scam limit," allowing highly diplomatic nations to siphon resources passively.
- **Migration/Expansion:** If a population exceeds 68% of its carrying capacity (`expansion_pressure_threshold`), it attempts to spill over into uninhabited neighboring land tiles. It prioritizes empty tiles with the highest carrying capacities and sends an overflow fraction of its population as migrants.

7. Visualization & UI Features

The simulation utilizes Pygame for an interactive, resizable visualization dashboard (`viewer.py`).

- **Interactive Camera:** Features panning, zooming (mouse wheel or +/- keys), and tracking capabilities.
- **Map Modes:** The user can toggle between different visual filters using the number keys (1-5) or 'M'.
  - *Terrain/Resources:* Shows land, water, and static resource density.
  - *Tech/Diplo/Physical:* Generates heatmaps visualizing the genetic spread of specific traits across the world.
  - *Lineage Coloring:* Custom RGB calculations assign dynamic brightness to lineage hex colors based on current population sizes relative to the global maximum.
- **HUD Overlay:** Displays live step counts, dominant traits, combat event tallies, dynamic tooltips (hovering over populations reveals exact counts), and attack arrows representing recent military actions.